

Arrays and ArrayLists

Java ArrayList

In Java, an `ArrayList` is used to represent a dynamic list. While Java arrays are fixed in size (the size cannot be modified), an `ArrayList` allows flexibility by being able to both add and remove elements.

```
// import the ArrayList package
import java.util.ArrayList;

// create an ArrayList called students
ArrayList<String> students = new
ArrayList<String>();
```

Index

An index refers to an element's position within an array. The index of an array starts from 0 and goes up to one less than the total length of the array.

```
int[] marks = {50, 55, 60, 70, 80};

System.out.println(marks[0]);
// Output: 50

System.out.println(marks[4]);
// Output: 80
```

Arrays

In Java, an array is used to store a list of elements of the same datatype. Arrays are fixed in size and their elements are ordered.

```
// Create an array of 5 int elements
int[] marks = {10, 20, 30, 40, 50};
```

Array creation in Java

In Java, an array can be created in the following ways:

- Using the `{ }` notation, by adding each element all at once.
- Using the `new` keyword, and assigning each position of the array individually.

```
int[] age = {20, 21, 30};
```

```
int[] marks = new int[3];  
marks[0] = 50;  
marks[1] = 70;  
marks[2] = 93;
```

Changing an Element Value

To change an element value, select the element via its index and use the assignment operator to set a new value.

```
int[] nums = {1, 2, 0, 4};  
// Change value at index 2  
nums[2] = 3;
```

Modifying ArrayLists in Java

An `ArrayList` can easily be modified using built in methods.

To add elements to an `ArrayList`, you use the `add()` method. The element that you want to add goes inside of the `()`.

To remove elements from an `ArrayList`, you use the `remove()` method. Inside the `()` you can specify the index of the element that you want to remove.

Alternatively, you can specify directly the element that you want to remove.

```
import java.util.ArrayList;

public class Students {
    public static void main(String[] args) {

        // create an ArrayList called
        studentList, which initially holds []
        ArrayList<String>
        studentList = new ArrayList<String>();

        // add students to the ArrayList
        studentList.add("John");
        studentList.add("Lily");
        studentList.add("Samantha");
        studentList.add("Tony");

        // remove John from the ArrayList,
        then Lily
        studentList.remove(0);
        studentList.remove("Lily");

        // studentList now holds [Samantha,
        Tony]

        System.out.println(studentList);
    }
}
```